



Society of Petroleum Engineers

SPE-229631-MS

The Sudair Formation: From Seal to Reservoir, Oman

L. Galluccio, Badley Ashton & Associates Ltd, Abu Dhabi, UAE; N. Al-Sinawi, A. Al Harthi, A. Al Hajri, G. A. Coy, W. Al Salmi, and Z. Al Rawahi, Petroleum Development Oman, Sultanate of Oman

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229631-MS](https://doi.org/10.2118/229631-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

The carbonate Sudair Formation (Lower Triassic) was originally considered as top seal of the oil and gas-bearing Khuff Formation due to the tight and argillaceous nature of its sediments. Nonetheless, the transition to grainier fabrics towards the north-east of Oman (platform margin) resulted in good reservoir facies, which are now the new focus play for exploration in Oman. This study provides a regional, integrated sedimentological, diagenetic and reservoir quality evaluation of the subsurface Sudair Formation, which has only been described previously in outcrop (Middle Mahil Member). The work has been performed at regional scale, across several fields, and provides an in-depth evaluation of the vertical and lateral facies distribution and related reservoir properties. This bridges an important knowledge gap in the region.

Detailed core sedimentological characterisation of four wells capturing the various members of the Sudair Formation (Lower, Middle and Upper) was carried out. Utilising core-to-log calibration, these wells were correlated with nine uncored wells to better constrain the stratigraphic architecture of the reservoir units. In-depth petrographic evaluation was performed on thin-sections and results were integrated with SEM, XRD, QEMSCAN analyses and available porosity/permeability measurements to unravel mineralogical composition and reservoir quality controls. Depositional maps were also used to enhance the characterisation of the reservoir in the inter-well areas by improving the spatial prediction of the rock properties.

The sedimentological architecture shows a series of stacked sequence stratigraphic cycles comprising intertidal (IT) and supratidal (NS) domains deepening upwards into high-energy inner ramp environments (IHE) – transgressive hemicycles, which then shallow upwards into intertidal and nearshore towards the cycle tops – regressive hemicycles. The comparison with neighbouring fields in the study area has highlighted that the formation underwent extensive erosion towards the east and south because of the Triassic/Jurassic regional unconformity. The truncation derived from this event caused the loss of the entire Jilh Formation and Upper Sudair Member in many of the eastern wells. Thin-section analysis highlights a complex diagenetic overprint and pervasively dolomitised facies. Dominantly mimetic dolomitisation preserved most of the sedimentary features and allochems of the precursor limestone. Overall, grainier fabrics present higher reservoir quality than mud-supported facies. Early and late dissolution phases

created macroporosity and enhanced microporosity, along with fracturing, which locally increased pore connectivity. Cementation is overall weak across the studied wells.

This core, log and thin-section-based integrated work provides key insights into the Sudair Formation. These allow the modelling and prediction of facies heterogeneity as well as identification of the best-connected sweet spots, which greatly help exploration and development strategies.

Introduction

Although previously described as a seal/barrier (Whittle and Alsharhan 1995), the Sudair Formation has been proven to have good reservoir quality within some fields. In particular, in north Oman, the formation represents a potential new focus play for exploration. In subsurface, this formation is both poorly documented and poorly understood due to the extremely limited availability of core data. Nonetheless, the recent acquisition of cores from PDO provided, for the first time, subsurface data covering different intervals of the Middle Sudair Member. This represents the topmost unit of the Sudair Formation in this area, where the progressive truncation towards the east completely eroded the Upper Sudair and the Jilh Formation.

Based on the above, the current study is crucial to support the development of a regional understanding of the facies distribution, diagenesis (including karstification associated with the basal Jurassic unconformity), controls on reservoir quality and intraformational seal development in the Sudair Formation. The work is focused on newly acquired cores capturing the basal Mafrak Formation shales (top seal for Sudair), the base Jurassic unconformity surface and the topmost interval of the Middle Sudair Member.

Geological background

The Sudair Formation was deposited as part of an epeiric carbonate platform during the Lower Triassic. The platform formed along the passive margin of the Neo-Tethys Ocean during the flooding of the Arabian Peninsula and generated a flat and consistent shallow water environment, where lateral facies changes were gradual and uniform (Osterloff et al. 2004; Pöppelreiter et al. 2011). This resulted in a consistent layer cake organisation of the sediments reflected in a consistent sequence stratigraphic architecture, where sequence boundaries and maximum flooding surfaces can be traced for kilometres (Sharland et al. 2001). The formation comprises an intercalation of shales, anhydrites and dolomite layers, with only limited occurrence of grain-supported elements within the generally tight dolomites (Alsharhan and Kendall 1986). A trend of decreasing shale and anhydrite towards the north-east, also associated with an increase in grain-supported fabrics within the dolomitised sediments, was also noted by Pöppelreiter et al. (2011). This corresponds to the seaward migration of the depositional belts highlighted by Ziegler (2001, Figure 1).

Whilst the clastic edge of the Sudair Formation constitutes an important aquitard and seal in Saudi Arabia and Syria (Alsharhan et al. 2001), in central Oman the carbonate Sudair Formation was originally considered as top seal of the underlying oil-bearing Khuff Formation, due to the tight and argillaceous nature of its sediments (Alsharhan and Nairn 1997). Nonetheless, the formation evolves into reservoir towards the north-eastward margin of the platform, where the transition to grainier fabrics is noted as the platform migrates seaward (Malek et al. 2005; Husain et al. 2008).

The Sudair reservoir is subdivided into six members, which comprise, from base to top: the Lower Sudair 1, Lower Sudair 2, Middle Sudair 1, Middle Sudair 2, Middle Sudair 3 and Upper Sudair 1. The Formation is capped at the top by a shale bed conformably overlain by the Jilh Formation. The latter is eroded away in the south and eastern part of the Oman Interior because of a regional unconformity (Al Sharhan et al. 1993, Obermaier and Aigner 2012 and references therein). This was caused by a major regression in the Late Triassic, due to epeirogenic movement and a westerly tilting of east Oman during Early Jurassic times, which led to sedimentary hiatuses between the Triassic and Jurassic (Alsharhan 1993). The prominent angular unconformity is regionally observable in seismic and illustrates an erosional truncation of the Permian-Triassic sediments and an eastward onlap of the Jurassic Mafrak Formation (Mountain and Prell 1990).

As a result, the complete Sudair Formation only occurs in the western and northern subsurface of the Oman interior and is truncated by Jurassic and Cretaceous units to the south and east (Forbes et al. 2010). The subsequent sedimentological make-up in these areas sees the Middle Sudair in direct contact with the overlying clastic Mafraq Formation (Mountain and Prell 1990).

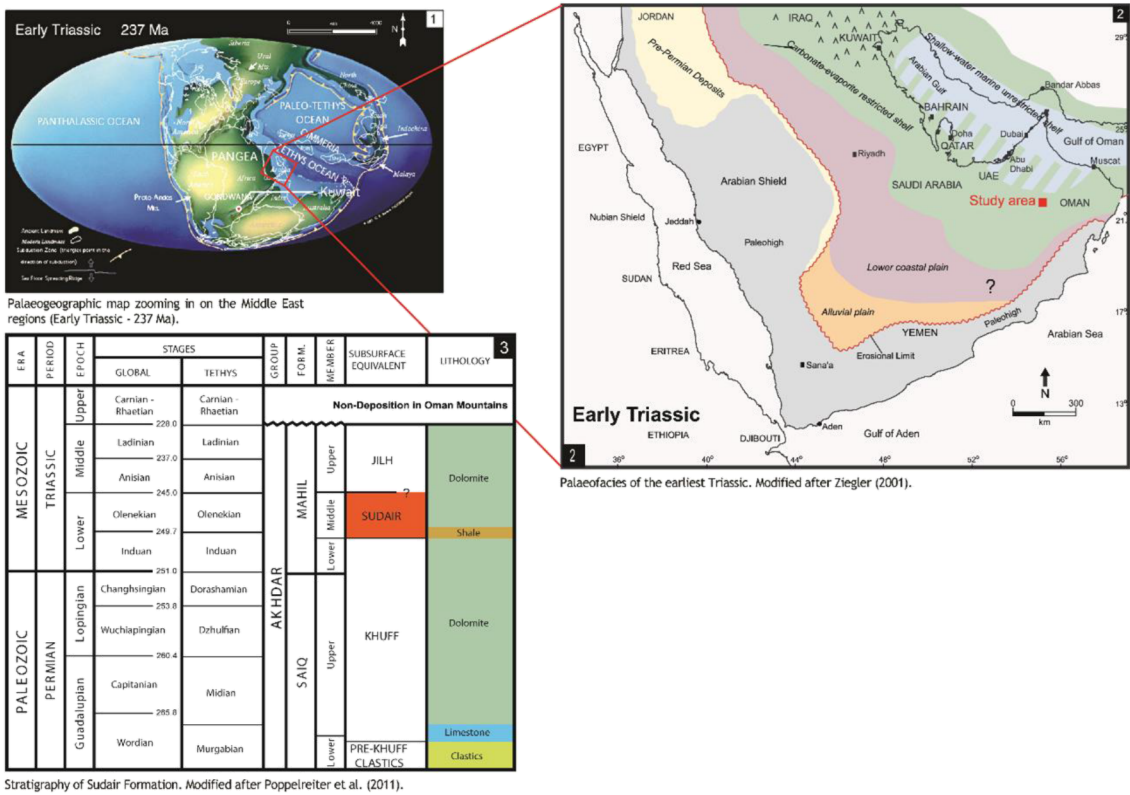


Figure 1—Early Triassic paleogeographic map (1); Location map of the study area on the regional gross depositional environment map, Arabian Gulf (2 – note that the meanings of the colours on the map are explained on the map itself); Stratigraphy of the Sudair Formation (3).

Material, methods, and workflow

The study is based on the sedimentological description of three partially cored wells that variably capture the Middle Sudair in two fields, and their correlation with ten other neighbouring wells/fields. A total of 78m of good quality slabbed cores were available for description. This allowed the characterisation of the Sudair/Mafraq unconformable boundary and provided depositional context with the Sudair Formation. The limited core coverage partially impairs a high-confident interpretation of the vertical and lateral facies evolution for the other members of the Sudair Formation across the study area. Core observations were integrated with a suite of openhole logs, including gamma ray, neutron, density, resistivity, sonic, porosity, lithology and hydrocarbon saturation. The variable log coverage locally reduces the confidence of core-to-log calibrations and interpretation of uncored intervals and correlation, especially in the Lower Sudair Member.

A data set of 153 horizontal and vertical core plugs was available from the three of the newly described wells. Conventional core analyses (CCA) were only successfully obtained for a subset of the samples (113 permeability and 118 porosity measurements). This is due to the highly brittle nature of the plugs, especially in the most cemented horizons, which resulted in failure during measurement. Out of these plugs, 32 were selected for thin-section petrography based on sedimentological characteristics, stratigraphic location and associated plug porosity and permeability values to ensure the best possible coverage in terms of vertical distribution and reservoir heterogeneity. The key sedimentological and diagenetic parameters for each thin-

section were qualitatively documented by detailed petrographic thin-section descriptions, whilst quantitative data on the cement abundance, pore types, skeletal and non-skeletal allochems and dolomitisation were also collected via modal analysis. These parameters were subsequently used to interrogate the CCA data to assess the reservoir quality heterogeneity within the Middle Sudair Member. Thin-section observations were also integrated with CCA, X-ray diffraction (XRD) and quantitative evaluation of minerals by scanning electron microscopy (QEMSCAN). A subset of 10 samples was also selected across the three wells for scanning electron microscopy (SEM) analysis aimed at characterising microcrystalline dolomite and microporosity.

Sedimentology

Lithofacies. The bed-scale lithofacies have been described using the classifications introduced by Dunham (1962) and Embry and Klován (1971). The scheme is texture-based and uses non-genetic descriptors of rock-fabric, mechanical and biogenic structures, with qualifiers denoting the occurrence of specific features (e.g. l for laminated and p peloidal). Eleven lithofacies have been recognised in the study wells, of which only four are present at an appreciable percentage to be considered relevant from a reservoir quality point of view. All the lithofacies are pervasively dolomitised by mimetic dolomite, which preserved the precursor limestone texture and sedimentary features. The observed lithofacies can be divided into two main groups: grain-supported and mud-supported fabrics.

The grain-supported fabrics comprise packstones, pack-grainstones and grainstones with a limited contribution of floatstones with grain-supported matrices, rudstones and breccias. These comprise ghosts of bivalves, benthic foraminifera and gastropods.

Mud-supported textures encompass mudstones, mud-wackestones, wackestones and wacke-packstones. These are normally barren or present a very limited faunal content, including ghosts of undifferentiated skeletal debris. These facies also locally contain microbial laminations and rootlet traces. Very rare siliciclastic mudrocks also occur intercalated within the dominantly carbonate succession.

Genetic elements. The Sudair Formation exhibits a diverse range of lithofacies, which account for different depositional textures and skeletal/non-skeletal allochems. These lithofacies display fine-scale variations that can be grouped into genetically related discrete units, referred to as genetic elements (GEs). These represent fundamental building blocks for understanding the reservoir architecture and variations in reservoir quality. The hierarchical approach used to interpret the GEs is based on Galluccio et al. (2022). The genetic elements recognised in the cored sections are listed below in order from the most proximal to more distal and displayed in table 1 for completeness.

Table 1—Genetic element scheme used in the present study.

Environmental belt and energy level											
Gross depositional setting		Dominant energy level/hydrodynamism		Environment	Code	Key environmental and depositional processes/characteristics (non-exhaustive listing)		Dominant carbonate texture	Dominant sediment structures	Allochems and Dominant trace fossils	
Supratidal	Exposed area above high tide mark	Variable		Supratidal	ST-NS-CH	Depending on climate, hydrology and/or coastal morphology evaporite successions may precipitate within salt or saline pans (ST). Paleosols may also develop, karst dissolution may occur (eg. humid conditions). The environment may also be sourced by continental siliciclastic input (sand/silt -CH- to clay -NS)		Mudstone, siliciclastic mudrocks, sandstone and siltstone	Moderate to high bioturbation, desiccation cracks, root traces	Undifferentiated bioclastic debris, possibly including bivalves. Root traces, together with <i>Teichichnus</i> and <i>Planolites</i> burrows.	
Intertidal	Area between low and high tide marks (frequently exposed)	Variable, but typically rather low		Intertidal	IT	Depending on climate may comprise salina/saksha (eg. promote microbial mat development and/or evaporite precipitation), coastal marshes and ponds where mangrove-like plants develop. May include high-energy tidal channels		Mudstone, mudstone-wackestone, rarely grainstone	Microbial lamination, rare bioturbation, root traces	Bivalve debris, which occur together with peloids and microbial lamination	
Subtidal	Inner/proximal (including lagoons where a barrier exists)	Above Fair Weather Wave Base (FWWB)	Low to moderate	Low to moderate-energy inner ramp	ILE	Typically the most proximal inner setting. Mainly characterised by mud-supported carbonate textures. Tidal channels may be present. This setting may include relatively shallow water lagoons if a well-developed hydrodynamic barrier exists (eg. oolitic shoal)		Mudstone, mudstone-wackestone, wackestone	Moderate to high bioturbation	Milolids, benthic foraminifera, algal clumps and bivalve debris, which occur together with peloids and <i>Thalassinoides</i> burrows	
			Moderate	Moderate energy inner ramp	IME	Typically influenced by wave action and tidal currents. May include the landward end of tidal inlets (if a hydrodynamic barrier exists) as well as build-up patches (eg. corals). Comprises washover deposition where sand sheets/bars, shoals, barrier islands, mounds exist and are reworked by storm events. This setting typically includes interbedded peritidal and periall environments (eg. backshoal, intershoal and/or foreshoal). Tidal inlets may be present in such settings if a hydrodynamic barrier is developed. Typically mud to grain-supported carbonate textures including interbedded wackestones, floatstones, packstones and subordinate grainstones		Wackestone-packstone, packstone	Moderate to high bioturbation	Ooids, oncoids, gastropods, echinoderm fragments, intraclasts, coated grains, bivalves, possible corals, benthic foraminifera and <i>Thalassinoides</i> burrows	
			High	High-energy inner ramp	IHE	Strong hydrodynamism dominated by waves and currents. Typically includes sand sheets/bars, shoals, barrier islands, mounds, reefs and/or build-ups if these are developed enough to create a hydrodynamic barrier (eg. shoal complex) the landward proximal side of the barrier (ie. former inner ramp) becomes a 'lagoon'. May experience exposure. Typically grain-supported textures		Packstone-grainstone, grainstone, floatstone and rudstone	Moderate bioturbation or planar laminations	Common ghosts of ooids, coated grains, peloids, oncoids, gastropods, bivalves, undifferentiated bioclasts, intraclasts, bivalve fragments	
	Middle/intermediate	Between FWWB and SWWB	Moderate	Moderate-energy Middle-ramp	IME	Typically open marine and well-oxygenated conditions where mud-supported carbonate textures dominate (wackestones to packstones)		Wackestones to packstones	Moderate to high bioturbation	Organic-rich, undifferentiated foraminifera, algal clumps, bioclasts, thin-shelled gastropods, bivalve and echinoderm fragments	
			Low	Low-energy Middle-ramp	MLE	Low-energy open marine and limited light penetration associated to a thicker water column in the mid-ramp environment					
	Outer/distal	Below Storm Weather Wave Base (SWWB)	Very low	Outer-ramp	OR	Typically very low-energy deposition from suspension settling of mud-dominated textures mainly resulting in a horizontally-laminated fabric. Typically organic-prone to organic-rich		Mudstone, wackestone	Low to moderate bioturbation	Organic-rich, rare foraminifera and undifferentiated bioclasts	

Supratidal genetic element: clastic nearshore (NS) is rare in the Sudair Formation. This is only observed in core in the Middle Sudair, where it comprises dm to m-thick beds of bioturbated (*Teichinchnus* and *Planolites*) siliciclastic mudrocks, where desiccation cracks and root traces are noted. Rare undifferentiated bioclastic debris is also documented, possibly including bivalves.

Intertidal genetic element (IT) occurs throughout the entire succession and is dominant in Sudair Middle 3. It generally consists of mudstones and wackestones with bird's eye structures, more rarely grainstones (locally oxidised), which locally contain microbial lamination, rare bioturbation and root traces. Bivalve debris is locally noted, together with peloids and anhydrite laths.

Low-energy inner ramp genetic element (ILE) is typically the most proximal subtidal setting, and is dominated by bioturbated, mud-supported textures (mudstones and wackestones), with only very rare occurrence of subordinate wacke-packstones. These deposits comprise minor to locally common benthic foraminifera (including miliolids and undifferentiated textulariids), bivalves, undifferentiated skeletal debris and green algae, alongside abundant peloids. This GE is episodically polluted by clay input from the land, which partially to completely shuts down the carbonate production, generating intervals with minor, monospecific faunal content (dominantly benthic foraminifera and bivalves). These are classified as argillaceous, low-energy inner ramp deposits (ILE.a).

Moderate-energy inner ramp genetic element (IME) is typically influenced by wave action and tidal currents. It is predominantly made up of moderately to highly bioturbated, packstone-dominated deposits containing common bivalve fragments and benthic foraminifera including undifferentiated textulariids, gastropods, bivalve and echinoderm fragments, as well as rare possible corals, common ooids, oncoids and coated grains. Moderate-energy deposits are often affected by continental run-off, which results in intervals typified by high clay content, identified with the code IME.a.

High-energy inner ramp genetic element (IHE) is characterised by the grainiest textures of the Sudair Formation as a result of strong hydrodynamism due to waves and currents. Pack-grainstones, floatstones with grain-supported matrices and rudstones dominate this GE. These are locally bioturbated and laminated and encompass an allochem assemblage including common ooids, coated grains, peloids, oncoids and gastropods. This is overall the most abundant GE recorded in the Sudair Formation.

Depositional Models. The depositional model interpreted for the pervasively dolomitised Sudair Formation (Figure 2) consists of broad facies belts that migrate landward/seaward based on sea-level fluctuations and local changes in accommodation space and hydrodynamism. The described facies and the allochem assemblages suggest an overall shallow marine setting subject to variable depositional energy, locally or episodically reaching permanent exposure.

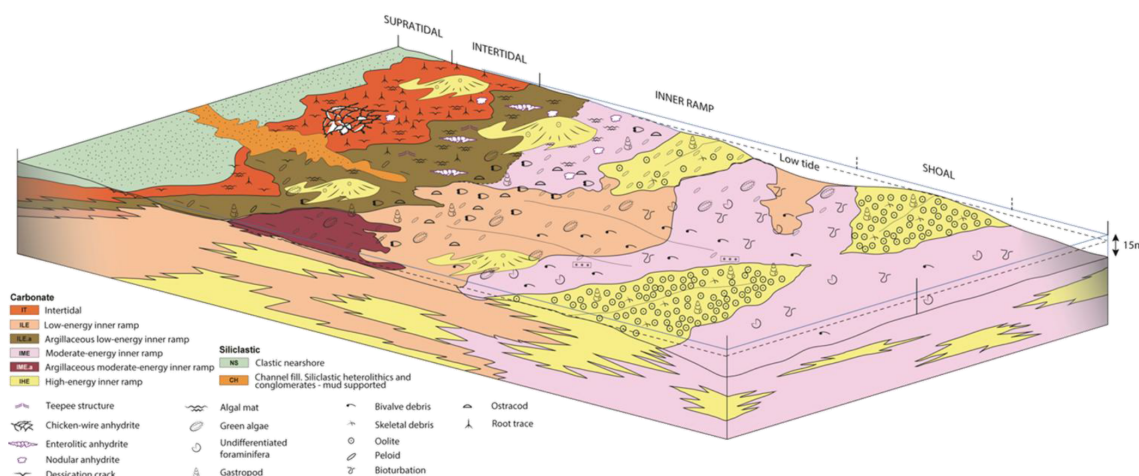


Figure 2—Conceptual depositional model for the Sudair Formation, onshore Oman.

The shallowest facies comprise siliciclastic mudrocks, generally poorly bioturbated and laminated, containing a rare and poorly diversified faunal assemblage, suggesting stressed environmental conditions. These facies, defined as nearshore deposits (NS), were deposited as a result of land-derived sediment influx in response to episodic increases in continental run-off, probably due to climatic perturbations. This environment typically interfingers with an intertidal, carbonate domain (IT), comprising microbialites and argillaceous mud-supported facies, which are characterised by a generally poorly diversified biota also reflecting environmentally stressed conditions (e.g. hypersaline waters), where faunal proliferation is impaired. Bird's eye structures, oxidised clays and grains are common, together with teepee structures, root traces, desiccation cracks and variably thick (cm to dm) brecciated interbeds, all pointing to a very shallow environment affected by episodic exposure influenced by tidal currents. These shallow sediments are located landward and are locally nested within subtidal deposits typical of an inner ramp setting, which are characterised by variable energy levels and clay input from the land (Figure 2). In more detail, argillaceous low (ILE.a) and moderate-energy (IME.a) inner ramp occupy the shallowest subtidal zone, where carbonate sedimentation is still polluted by clay input from the nearby intertidal and nearshore areas. These are intercalated with more abundant, cleaner, low and moderate-energy inner ramp (ILE and IME), which show a more diversified floral-faunal assemblage, with the occurrence of common algae and benthic foraminifera, indicative of more "normal" marine conditions. The transition from low to moderate energetic conditions is marked by a dominance of mud vs grain-supported fabrics, with mud-supported (e.g. mud-wackestones) depicting a low-energy and grain-supported (e.g. packstones) representing a moderate-energy domain. Development of high-energy sediments is also noted within the inner ramp environment. These are made of peloidal pack-grainstones and grainstones commonly containing reworked bivalves and gastropods, as well as abundant peloids and ooids. These can form shoals or sand bars that are likely to be very sensitive to minor topographic variations and develop primarily over palaeo-highs. Rare high-energy interbeds can also occur interfingered in lower energy deposits as thin horizons. These are indicative of potential small patches and sand sheets rather than continuous shoals (Figure 2).

Sequence stratigraphic framework. The sequence stratigraphic framework established for the Sudair Formation in the study area is based on the described sedimentological trends, as well as regional understanding from literature (Whittle and Alsharhan, 1995; Sharland et al. 2001; Forbes et al. 2010; Pöppelreiter et al. 2011; Obermaier et al. 2010). An overview of the vertical and lateral evolution of the Sudair Formation in light of its sequence stratigraphic framework is displayed in Figure 3.

Overall, the Sudair Formation describes a complete 2nd-order cycle in which four 3rd-order cycles and eight 4th-order cycles are nested. These are bounded at the top by argillaceous beds in conformable contact with the overlying Jilh Formation in the western part of the study area, whilst the contact becomes erosional towards the east and south, due to a regional unconformity that brings the Jurassic Mafraq Formation in direct contact with the Triassic Middle Sudair. This unconformity (SB-J10) is also reported in literature (Alsharhan et al. 1993, Obermaier and Aigner 2012, and references therein). The erosion determines the truncation of many of the mentioned sequences, such that in the truncated areas, the overall thickness of the Sudair Formation can reach half of the thickness seen in the west, where the formation is complete. The formation presents a well-developed layer-cake facies architecture, with only minor variations in thickness and facies make-up from one well to another. Vertically the formation depicts a 2nd-order cycle where proximal environment varying from intertidal setting (IT), intercalated with argillaceous, low- to moderate-energy inner ramp settings (ILE and IME.a, respectively) to slightly deeper, moderate- (IME) and high-energy inner ramp (IHE) sedimentation. The maximum flooding zone is placed within the thickest accumulation of grain-supported sediments, whilst tight, highly cemented and argillaceous deposits occur mostly at the base of transgressive phases (Figure 3).

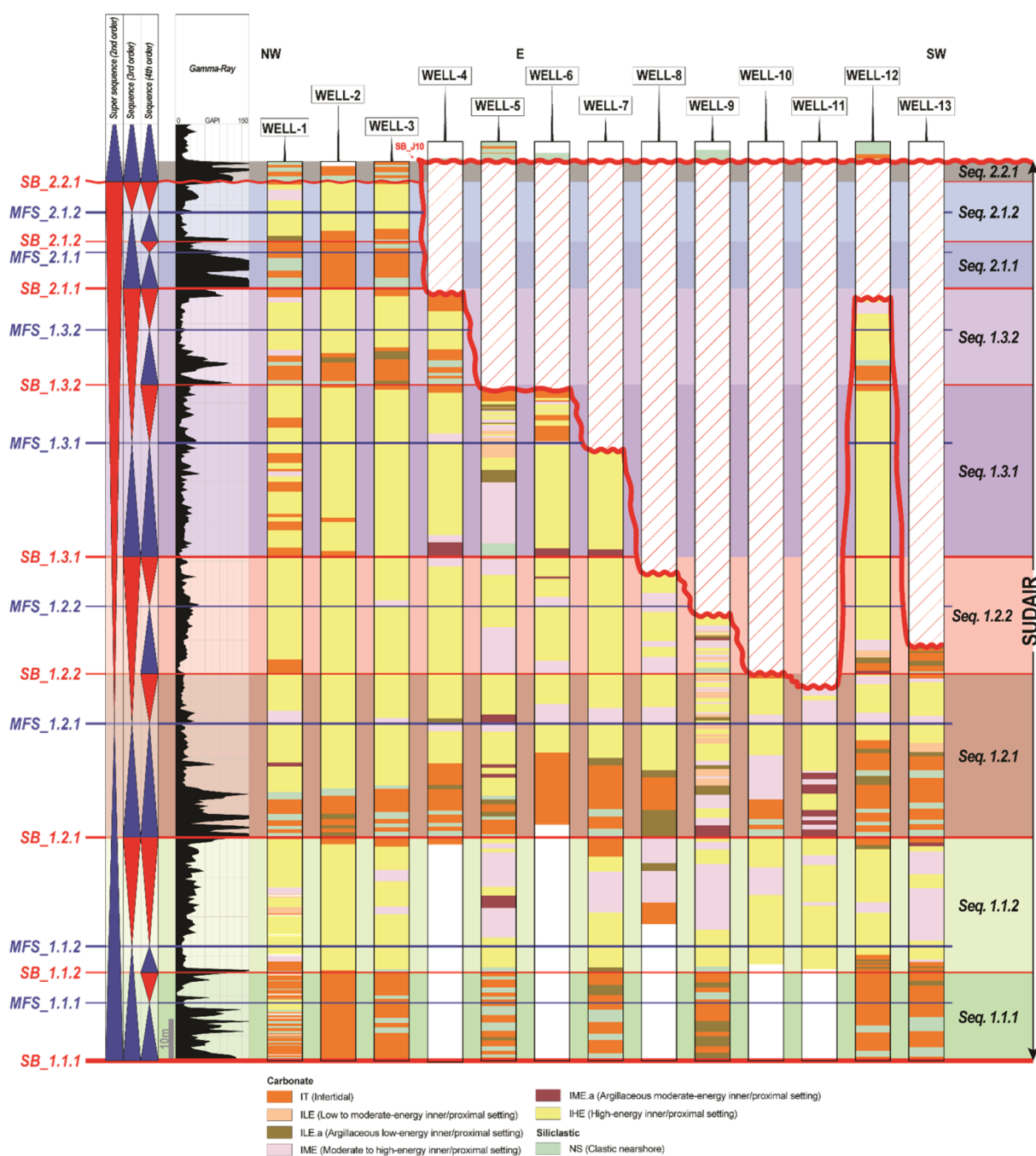


Figure 3—Correlation panel of the Sudair Formation, onshore Oman.

The identified 3rd and 4th-order cycles broadly present a very similar vertical evolution. The transgressive hemicycle is generally characterised by an upward-deepening from intertidal (IT) and nearshore deposits (NS) to relatively more distal high-energy inner ramp deposits (IHE), with the overlying regressive hemicycle describing the opposite motive (IHE returning to IT). Typically, sequence boundaries (SB) are picked at the base of the shallowest GEs (IT, NS), whilst maximum flooding surfaces (MFSs) are positioned within the thickest accumulation of relatively deeper IHE and IME deposits (Figure 3).

Whilst the sequences associated with the Lower Sudair and the Middle Sudair 1 are considered continuous across the correlated wells, the sequences capturing the Upper Sudair and Middle Sudair 2 and 3 are variably truncated by the SB-J10 unconformity across the study area. In particular, the 4th-order sequence comprising the Middle Sudair 3 and part of the Middle Sudair 2 are complete only in the westernmost wells, whilst

they are truncated, with different degrees of intensity, and are completely absent in wells 9, 10 and 11. The topmost part of the Sudair Formation (Middle Sudair 1, Upper Sudair 1 and 2 and base Jilh Formation) is only captured in the westernmost wells, whilst it is completely truncated elsewhere by the top Triassic/Jurassic regional unconformity (SB_J10; [Al-Husseini 2009](#)).

At regional scale, the comparison between the study fields highlights that the Sudair Formation shows a gross trend in reduction of clastic input (NS) and transition to more energetic, high-energy inner ramp sediments (IHE) towards the east. This is in line with the north-eastward deepening of the carbonate platform reported in literature and the consequent increase in grain-supported fabrics with increasing water depth and agitation ([Ziegler, 2001](#); [Forbes et al. 2010](#); [Pöppelreiter et al. 2011](#)).

Depositional trend maps. The Sudair Formation areal depositional trends are reported in the maps displayed in [Figure 4](#). Eight sets of genetic elements and log porosity maps were created based on the eight 4th-order sequences identified, and an extra map was generated to capture the basal part of the transgressive hemicycle depicting the topmost argillaceous interval of the Sudair Formation, directly underlying the Jilh Formation.

The basal three 4th-order sequences (1.1.1, 1.1.2, 1.2.1, [Figure 4](#)) cover the entire area of interest with only a small erosion of sequence 1.2.1 noted in Well-11. The basal map (sequence 1.1.1, [Figure 4](#)) shows the dominance of intertidal (IT) and nearshore (NS) domains at the base of the succession with IT more abundant in the west/south-west of the study area, with an increase in siliciclastic input noted towards the central and eastern regions. Argillaceous inner ramp setting (ILE.a) becomes more prominent in Well-9, possibly indicating a subtle north-eastward deepening. This GE is also apparently characterised by slightly better porosity than the other GEs in this time slice (sequence 1.1.1, [Figure 4](#)). Sequence 1.1.2 shows the transition to a dominant subtidal setting where high-energy deposits prevail, especially towards the north-west. In contrast, a slight reduction in energy levels is recorded towards the east, where moderate-energy domains prevail (IME, sequence 1.1.1, [Figure 4](#)). The log porosity map shows a subtle increase in porosity matching wells dominated by IHE. In sequence 1.2.1 ([Figure 4](#)) the incursion of more proximal deposits is noted, with a subtle northward transition from intertidal to subtidal settings. Within the subtidal domain, high-energy sedimentation dominates, especially towards the north-west (IHE), whilst low-energy environments (IME, ILE) are noted towards the east. The log porosity map shows an increase in porosity towards the south.

From sequence 1.2.2 upwards the effects of the truncation are noticeable, with lack of deposits in the south-eastern part of the area of interest. The region affected by erosion widens upwards culminating in sequence 2.2.1, which is solely captured in the westernmost wells. In terms of GE, all the truncated sequences, except for 2.2.1 and 2.1.1, which are dominated by intertidal (IT) and argillaceous nearshore domain (NS), show a dominance of high-energy inner ramp environment (IHE) generally passing to moderate-energy subtidal domain (IME) towards the central area and intertidal (IT) towards the west (e.g. sequence 1.3.1). Generally, porosity is moderate to good in correspondence with IHE and IME, whilst it becomes poor to extremely poor when NS dominates.

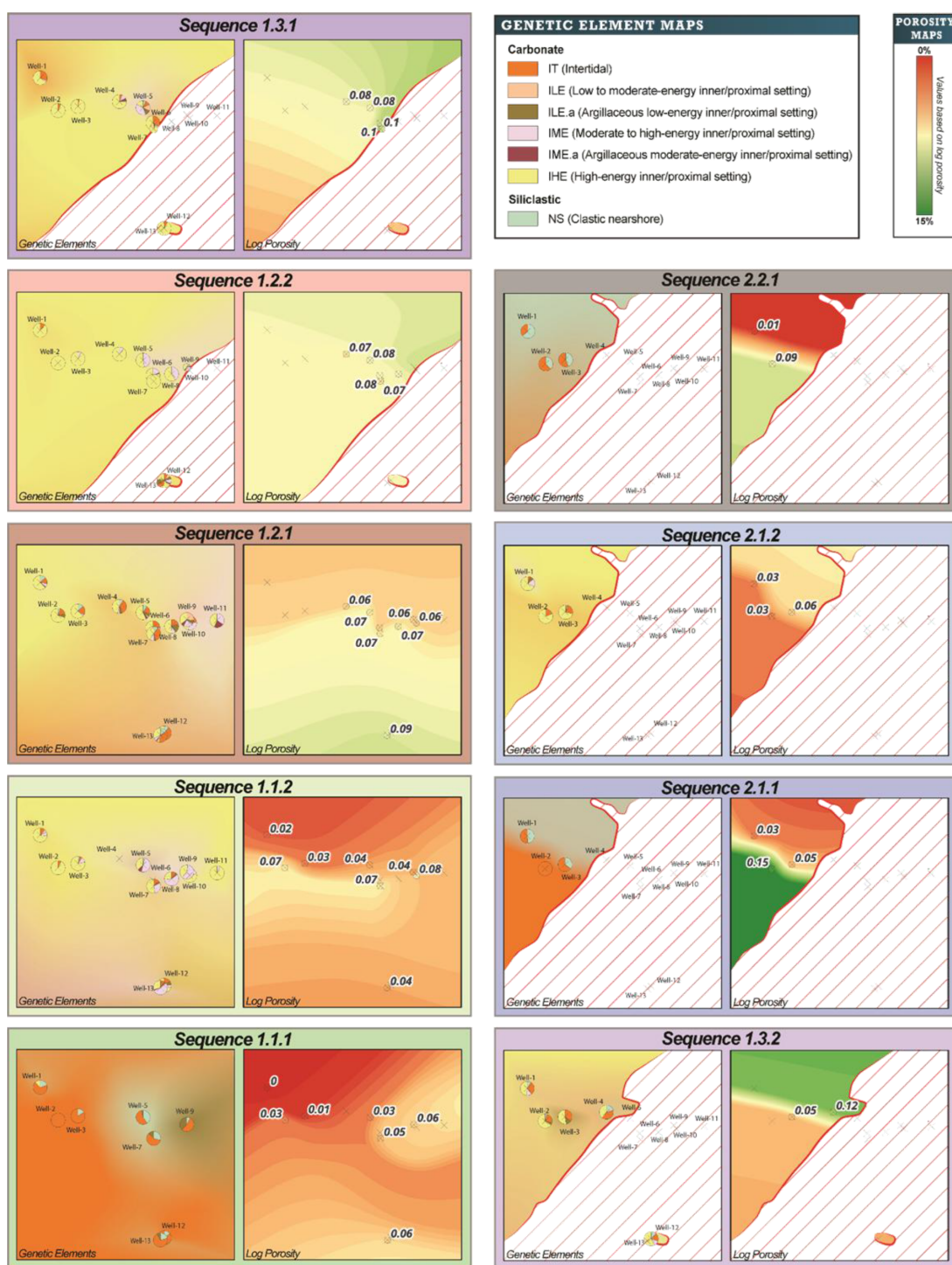


Figure 4—Depositional and log porosity maps for the Sudair Formation, onshore Oman.

Characterisation of diagenetic phases

Replacive dolomite. The entire Sudair Formation has been pervasively replaced by fine to medium-crystalline dolomite and microcrystalline dolomite. These present two different styles of dolomitisation, allowing, or not, preservation of the limestone precursor's original fabric and sedimentary features. Fabric retentive dolomitisation is the most common style of dolomitisation and results in the original texture and allochems remaining easily identifiable as ghosts. These are visible due to variations in either dolomite crystal size and packing or inclusion abundance within the dolomite crystals that generate darkened

haloes mimicking the precursor's original components. This style of dolomitisation is observed across all the facies, to the point that dolomitised high-energy grainstones and pack-grainstones are not only identifiable in terms of texture, but also in terms of their specific constituents (e.g. ooids, peloids, type of benthic foraminifera). This dolomitisation style allows full preservation of the original precursor's porosity (e.g. interparticle macropores between the grains), as well as the development of minor to common intercrystalline macropores and micropores in-between the dolomite crystals, and minor mouldic macropores and vugs. The best preservation of precursor features is seen when microcrystalline dolomite dominates pore-occluding phases. Overall, fabric-destructive dolomitisation is very rare and solely observed in one well (Figure 5 photomicrograph b). During this dolomitisation the original depositional texture is completely replaced with medium-crystalline anhedral dolomite, which generates a compact mosaic with no remnants of the precursor limestone texture. Development of intercrystalline macropores in-between the dolomite crystals is rare.

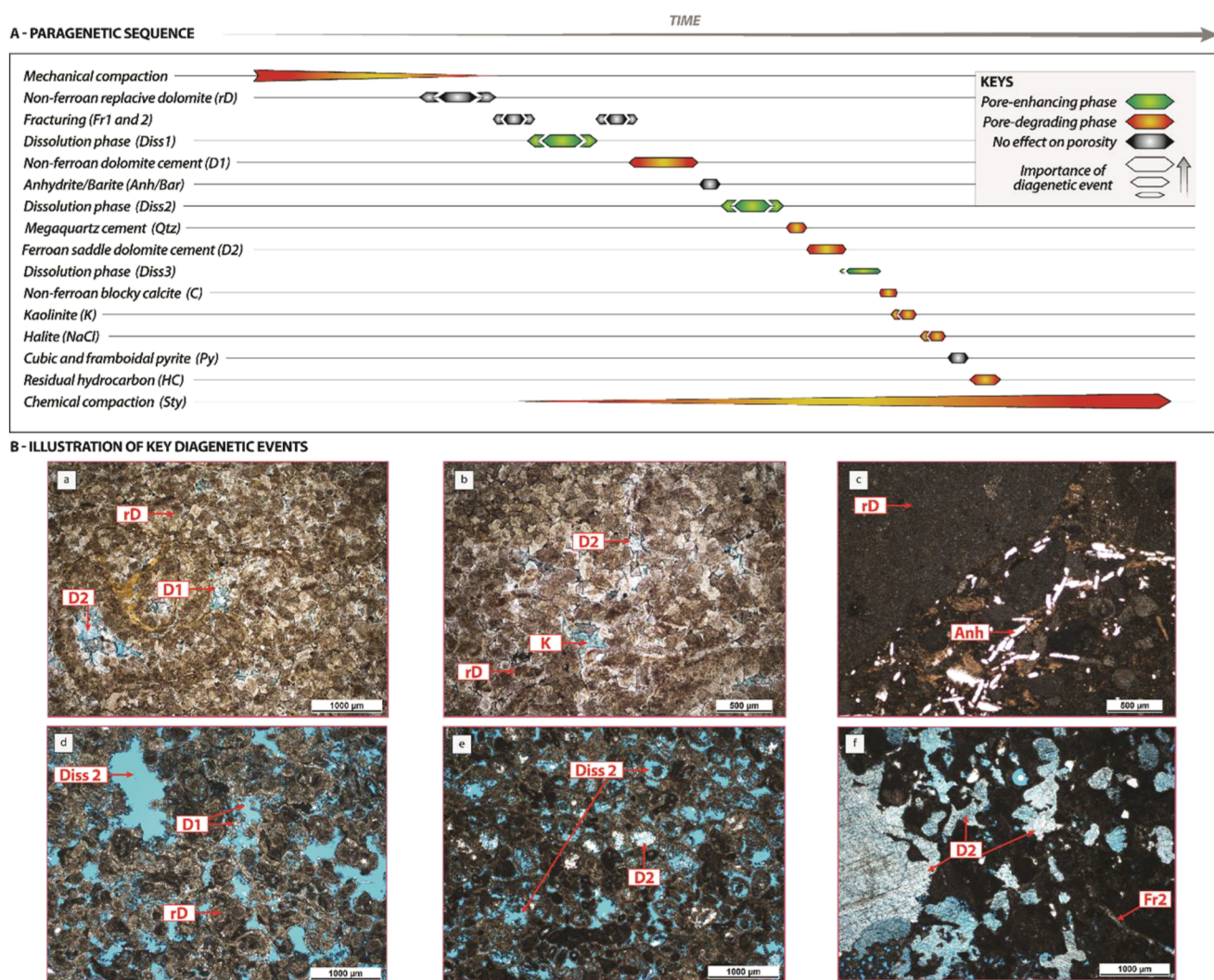


Figure 5—Paragenetic sequence of the Sudair Formation (A), and illustration of key diagenetic events (B).

Modal analysis on the three study wells highlights that there is a change in dolomite crystal size moving from the west to the east of the study area. The evaluation of the dolomite stoichiometry from XRD data (Lumsden and Chimahusky 1980) indicates that both dolomite crystal sizes present a composition very close to stoichiometry (close to 50% Mg/Ca ratio in the crystal lattice), with only a very subtle decimal difference

between the two. In more detail, microcrystalline dolomite is dominantly closer to stoichiometry, whilst the coarser dolomite has a slightly higher Mg/Ca ratio.

The evidence above may indicate that microcrystalline dolomite may be potentially slightly earlier than medium-crystalline dolomite. Therefore, the western wells may have been dolomitised slightly earlier than the eastern wells, where dolomitisation could have happened somewhat later. Considering the subtly deeper conditions recorded for the Sudair Formation in the eastern wells compared to the western wells (Ziegler, 2001), this could infer potentially sabkha-type dolomite formation in the west (where IT also dominates - highly evaporitic pore fluids, Cantrell et.al. (2004)) evolving to a reflux dolomitisation towards the east, where subtidal environments are more prominent than intertidal environments. The reflux process would be triggered by dense, saline waters moving downward from the evaporative area to deeper areas of the platform (Morad et.al. 2012; Saller and Henderson, 1998; McKenzie et al. 1980). Nonetheless, this interpreted dolomitisation model needs validation via a specific stable isotope study (carbon and oxygen) to understand the nature and origin of the fluids responsible for the dolomitisation process.

Based on the above, the impact of dolomitisation in reducing and/or enhancing reservoir quality is very limited, due to its dominant mimetic character, which makes the precursor limestone texture the key element driving diagenesis and pore evolution.

Pore-enhancing phases. The key pore-enhancing mechanism is dissolution, which creates/enhances macroporosity and microporosity. The most important dissolution event positively affecting reservoir quality is late dissolution (Figure 5A, Diss2, and Diss3). This is reflected by abundant enlarged interparticle macropores, vuggy porosity, secondary intraparticle macropores and rare moulds (Figure 5, photomicrographs d and e). These phases are testified by corroded early and late cement rims, as well as replacive dolomite edges and are likely to occur before and/or contemporaneously with oil emplacement (Wright and Harris 2013). Early dissolution (Diss1, Figure 5) has a minor effect in enhancing pore volume and connectivity. This process mostly affects replacive dolomite crystals, which appear corroded, especially when rimming macropores. This generally results in a subtle increase in intercrystalline macroporosity and microporosity. The fluids generating this dissolution are likely to be mixed meteoric/marine in origin and potentially occurring during or in proximity of exposure (Hollis 2011).

Fractures are a secondary pore-enhancing phase that can locally facilitate fluid flow. Two generations of fractures have been identified in thin-section. Their impact on the reservoir quality is considered limited and localised, based on the available dataset.

Pore-occluding phases. Several cementing, pore-occluding phases and compaction events that reduce pore volume and connectivity have been identified (Figure 5A and B). Out of the cementing phases two are key, whereas the others are minor, and therefore their negative effect on the pore volume and connectivity is limited and localised.

The first major cementing phase is the pore-filling, medium to coarse-crystalline, ferroan saddle dolomite cement (Figure 5A, D2). This is minor to common in abundance and occurs in both primary and secondary macropores and fractures and in a wide range of textures from mudstones to grainstones. It mostly fills macropores during burial diagenesis, partially to completely destroying the positive effect of early and late dissolution in enhancing pore volume and connectivity (Figure 5B, photomicrograph f).

The second most important pore-occluding phase is fine to rarely medium-crystalline, non-ferroan euhedral, inclusion-poor to inclusion-prone dolomite cement (Figure 5A, D1, Figure 5B, photomicrographs a and d). This is generally common to abundant and is also observed as overgrowths around replacive dolomite crystals. The phase lines interparticle and intercrystalline macropores reducing the pore-throat radii, but not completely occluding macropores. As a result, its negative impact on pore volume and connectivity is less important than D2. This phase is thought to be related to earlier burial diagenesis as it is enclosed by D2 and other later diagenetic phases.

Minor pore-occluding phases have a limited impact on the pore system, especially if taken separately, however, cumulatively they have a visible effect in reducing the total pore volume and connectivity. These phases include fine to medium-crystalline, non-ferroan blocky calcite (C), pore-occluding kaolinite (K in [Figure 5A](#) and, [Figure 5](#), photomicrograph b), halite (NaCl) and megaquartz (Qtz). Residual hydrocarbon is also present as a rare phase (HC), which partially occludes macropores.

Following cementation, mechanical compaction is seen to have an important impact in reducing pore volume and connectivity, especially in grain-supported textures across the three study wells. This process is likely to have occurred during the early stages of diagenesis, before the dolomitisation process and is inferred by point and planar to concavo-convex grain contacts that, with an increase in compaction and burial depth, rarely evolve to sutured grain contacts. Mechanical compaction is thought to have occurred before, contemporaneously to and after the formation of early marine calcite cements that are not visible in the present-day samples due to pervasive dolomitisation. Mechanical compaction that has post-dated the formation of early calcite cements is recognised by the spalling or breaking away of dolomite-replaced, fibrous calcite cement fringes from allochems. Chemical compaction is also interpreted to have occurred during the burial stages of diagenesis creating stylolites. The stylolites often concentrate clays, organic matter and pyrite along their planes. The latter is also scattered throughout the dolomite crystals.

Reservoir quality controls

Sedimentological controls. The texture of the original limestone precursor is the main control on the reservoir quality in the Sudair Formation. Textural changes from mud to grain-supported fabrics are responsible for a net increase in both porosity and permeability. This is clearly visible in [Figure 6](#), where the increase in porosity and permeability is the result of a higher proportion of connected interparticle macropores within the coarser grained textures (Trend 1). In fact, grain-supported fabrics, deposited in high-energy settings (IHE), favour the development of large and effective interparticle macropores (and pore throat radii). In contrast, more micrite-rich mudstones, wackestones and packstones, deposited under low and moderate-energy conditions (IT, ILE, IME), exhibit a reduced quantity and size of primary macropores (smaller pore-throat radii) and prevalence of intercrystalline micropores, generating a tortuous fluid flow.

Within grain-supported fabrics, grain-size is also a significant controlling parameter on permeability variations for a given porosity. [Figure 6](#) (Trend 3) exemplifies how coarser grained grainstones and floatstones with grainstone matrices, deposited in high-energy settings (IHE), are characterised by permeabilities about one order of magnitude higher than finer grained grainstones. The passage from fine to coarse-grained peloids, ooids and intraclasts leads to the formation of larger pore-throat radii, which generate a more effective connectivity ([Lucia et al. 2001](#)).

Sorting/packing also influences the reservoir quality in grainstones. In this study, a gross increase in permeability (about 1 to 2 orders of magnitude) and a less evident increase in porosity are noted passing from poorly to moderately sorted grainstones to well sorted grainstones. This is associated with variation in pore-throat radii size, which is larger in well-sorted compared to poorly sorted fabrics.

Clay content favours compaction and cementation, decreasing both macroporosity and microporosity. In the Sudair Formation, SEM observations highlight the presence of illitic clays that envelope microcrystalline dolomite reducing the pore-throat radii. Locally, the presence of clays also triggers development of tighter microcrystalline dolomite, due to enhanced cementation. Clay content is seen to be more abundant towards the shallowest sections of the carbonate ramp where the Sudair Formation deposited. This indicates that more proximal settings (NS, IT, ILE.a and IME.a), often associated with sequence boundaries, are expected to have lower reservoir quality than the other environments.

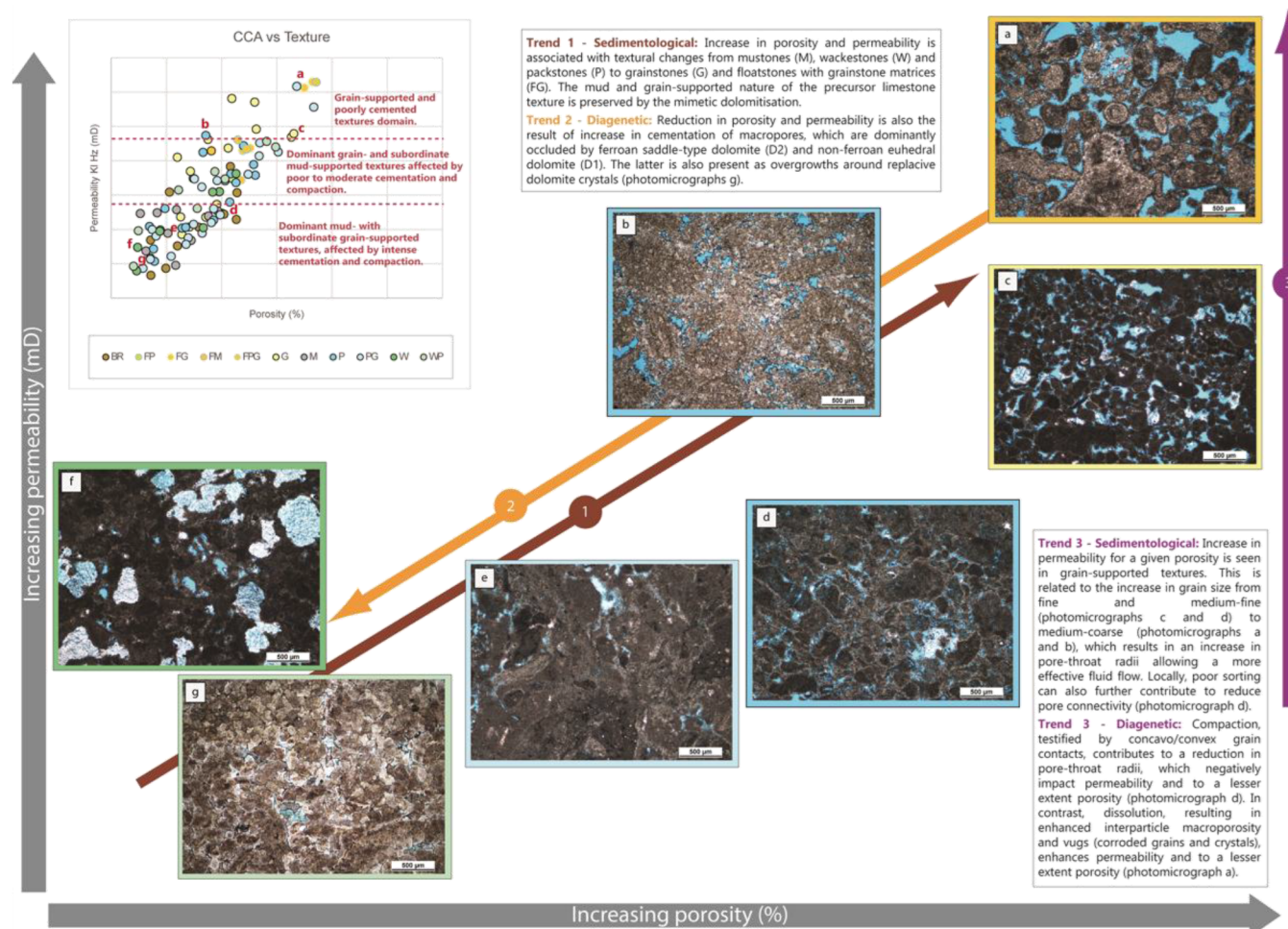


Figure 6—Integrated reservoir quality controls of the Sudair Formation, Onshore Oman.

Diagenetic controls. Cementation has an important negative effect on the reservoir potential as cements partially to completely occlude porosity and impair connectivity. In more detail, cementation leads to a decrease in both porosity and permeability as depicted in Figure 6 (Trend 2). Ferroan, saddle dolomite (Figure 5AD2) is the main cementing phase occluding macropores and it has the major detrimental effect on reservoir quality (Figure 6A, photomicrograph f), whilst non-ferroan, euhedral dolomite (Figure 5A D1) has a more limited impact on porosity and permeability as this mostly lines macropores, reducing pore-throat size. Overall, it is important to emphasise that the Sudair Formation is generally only moderately cemented, with rare cases of more intense cementation. This results in cementation being only a secondary control on the reservoir quality, due to its localised negative impact.

Mechanical compaction has a fairly negative effect on the reservoir quality of the grainstones. This determines a reduction of about one order of magnitude in permeability for a given porosity as exemplified by Figure 6 (Trend 3). This causes breakage and spalling of the ooids and peloids, as well as punctate and concavo/convex grain contacts that are all elements resulting in an overall reduction of pore-throat size or complete destruction of microporosity. The degree of compaction could be a factor linked to depth; however, early development of grain-rimming calcite cements could have, in some instances, also limited the impact of compaction with burial, due to the creation of a rigid framework preventing further reduction in pore spacing.

Dissolution events (especially Figure 5A Diss2 and Diss3; Figure 5B photomicrographs d and e) are seen as the key diagenetic phases that generate/enhance both macropores and micropores by dissolving the replacive dolomite alongside cements.

Dolomitisation has only a limited impact on the reservoir quality variations in the Sudair Formation, despite its pervasive nature. In fact, mimetic replacement has allowed the preservation of the majority of the original limestone precursor's pore system and effectiveness. Overall, only a subtle link between dolomite crystal habit and reservoir quality has been identified via modal analysis. In particular, dolomite mosaic type (planar-e vs planar-s and non-planar) seems to impact porosity and permeability, with better performing dolomites associated with the dominance of euhedral crystals vs anhedral interlocking crystals. This is in agreement with published literature (Galluccio et al. 2008, Galluccio 2010, Iannace et al. 2011, and references therein).

Conclusions

The Sudair Formation in the study wells exhibits an overall "layer-cake" organisation of the sediments, attributable to the very low gradient of the carbonate platform that characterised onshore Oman during the Triassic. This interpretation is supported by the observed sedimentological architecture. Comparison with adjacent fields within the study area reveals that the formation experienced significant erosion towards the east and south, driven by the regional Triassic–Jurassic unconformity. This truncation event resulted in the complete removal of the Jilh Formation and the Upper Sudair Member in many of the eastern correlation wells. In the most severely eroded sections, the late Triassic –lower Jurassic Mafrq Formation directly overlies the lower Triassic Middle Sudair 1 sequence.

From a diagenetic perspective, the selected samples display a complex diagenetic overprint, dominated by pervasive, mimetic dolomitisation. This process has preserved the majority of the sedimentary structures and allochems of the precursor limestone. As a result, reservoir quality appears to be minimally affected by dolomitisation. Conversely, the texture of the original limestone exerts a more significant influence on reservoir performance, with grain-supported fabrics demonstrating superior reservoir quality compared to mud-supported facies. Overall, the best reservoir properties are found within grainstones, packstones, and floatstones deposited in a high-energy inner ramp setting. These facies exhibit good to excellent permeability, attributed to a pore system dominated by large interparticle macropores, which are further enhanced locally by late-stage dissolution. In contrast, the lowest porosity and permeability values are associated with mud-supported, clay-rich deposits formed in proximal nearshore, intertidal, and argillaceous subtidal environments.

The findings of this study address a significant gap in knowledge and publications regarding the understanding of the Sudair Formation in the subsurface. A comprehensive characterisation of this formation provides a valuable tool for modelling and predicting subsurface heterogeneity within the Triassic of Oman, particularly in the context of tectonic evolution and associated erosion. This, in turn, will greatly enhance the ability to predict the most prospective lithofacies within the formation, supporting future exploration and production efforts.

Acknowledgements

We gratefully acknowledge the permission and support provided by PDO (Petroleum Development Oman) management for the publication of this article. Special thanks are extended to the PDO Exploration team for their valuable comments.

References

- Al-Husseini, M. 2009. Update to Late Triassic – Jurassic stratigraphy of Saudi Arabia for the Middle East Geologic Time Scale. *GeoArabia* 2009; p. 145–186, v.14(2). doi: <https://doi.org/10.2113/geoarabia1402145>
- Alsharhan, A.S. and C.G.ST.C. Kendall 1986. Precambrian to Jurassic rocks of Arabian Gulf and adjacent areas: Their facies, depositional setting and hydrocarbon habitat. *American Association of Petroleum Geologists Bulletin*, v. 70, no. 8, p. 977–1002.

- Alsharhan, A.S., A.E.M. Nairn, and A.A. Mohammed 1993. Late Palaeozoic glacial sediments of the southern Arabian Peninsula: Their lithofacies and hydrocarbon potential. *Marine and Petroleum Geology*, v. **10**, p. 71–78.
- Alsharhan, A.S. and A.E.M. Nairn 1997. Sedimentary basins and petroleum geology of the Middle East. Elsevier, Amsterdam, New York, Oxford, 843 p.
- Alsharhan, A.S., Z.A. Rizk and A.E.M. Nairn 2001. Hydrogeology of an Arid Region: The Arabian Gulf and Adjoining Areas. Elsevier, 325 p.
- Cantrell, D., Swart, P. and Hagerty, R., 2004. Genesis and Characterization of Dolomite, Arab-D Reservoir, Ghawar Field, Saudi Arabia. *GeoArabia*, **9**(2), pp.11–36.
- Dunham 1962. Classification of Carbonate Rocks According to Depositional Texture. In W.E. Hamm (Ed.). Classification of Carbonate Rocks, A Symposium. American Association of Petroleum Geologists, pp. 108–121.
- Embry, A.F. and Klován, J.E. 1971. A Late Devonian reef tract on Northeastern Banks Island, NWT. *Canadian Petroleum Geology Bulletin*, vol. **19**, pp. 730–781.
- Forbes, G.A., Jansen, H.S.M. and Schreurs, J. 2010. Lexicon of Oman Subsurface Stratigraphy. *GeoArabia Special Publication 5*.
- Forbes, G.A., Jansen, H.S.M. and Schreurs, J. 2010. Lexicon of Oman Subsurface Stratigraphy. *GeoArabia Special Publication 5*.
- Galluccio, L., Frijia, G., Iannace, A., Mazzoli, S., Parente, M., Vitale, S., Giorgioni, M. and D'Amore, M., 2008. Diagenesis and petrophysics of dolomite in the "Middle" Cretaceous of the Sorrento Peninsula (Southern Apennines). *Rendiconti Online Società Geologica Italiana* v.2, p. 87–92.
- Galluccio, L., 2010. Dolomitization within the Mesozoic carbonates of Sorrento Peninsula (Southern Apennines — Italy): genetic models and reservoir implications. Unpublished PhD Thesis available at www.fedoa.unina.it/4114/1/Galluccio_Laura_PhD_Thesis.pdf.
- Galluccio, L., Breislin, C. and Tendil, A.J.B., 2022, October. Capturing Reservoir Quality Heterogeneities at the Stage of Sedimentological Coding: An Innovative Depositional Environment Scheme. In Abu Dhabi International Petroleum Exhibition and Conference (p. D011S022R003). SPE.
- Husain, R., A.A. Sajer, N. Al-Zabout, A. Rabie, M.K. Iqbal and D.A. Khan 2008. Tectonic controls on Triassic stratigraphy and hydrocarbon prospectivity in Kuwait. 8th Middle East Geosciences Conference, GEO 2008. *GeoArabia*, Abstract, v. **13**, no. 1, p. 175.
- Iannace, A., Capuano, M. and Galluccio L., 2011. "Dolomites and dolomites" in *Mesozoic platform carbonates of the Southern Apennines: Geometric distribution, petrography and geochemistry*. *Palaeogeography, Palaeoclimatology, Palaeoecology*, v.**310**, p324–339.
- Lucia, F.J., Jennings, J.W., Rahnis, M. and Meyer, F.O. 2001. Permeability and Rock Fabric from Wireline Logs, Arab-D Reservoir, Ghawar Field, Saudi Arabia. *Geoarabia* **6**, 4, pp.619–646.
- Lumsden, D.N. and Chimahusky, J.S., 1980. Relationship between dolomite nonstoichiometry and carbonate facies parameters. *Soc. Ec. Pal. And Min. Spec. Publ.*, v. **28**, p.123–137.
- Malek, S., S.K. Abdul, R. Bhattacharya, A.A. Hussain, T. Sajer, K.O. Hearn and L. Hoi 2005. Gas-condensate discovery from the Sudair Formation of Kuwait exploration: Implications for Triassic Plays. American Association of Petroleum Geologists International Conference Paris. Abstract volume.
- McKenzie, J.A., Hsu, K.J. and Schneider, J.F. 1980. Movement of subsurface waters under the sabkha, Abu Dhabi, UAE, and its relation to evaporative dolomite genesis. In: Zenger, D.H., Dunham, J.B. and Ethington, R.L. (eds). *Concepts and models of dolomitization*. *SEPM Special Publication*, v.**28**, pp.11–30.
- Morad, S., Al-Aasm, I., Nader, F., Ceriani, A., Gasparrini, M., and Mansurbeg, H., 2012. Impact of diagenesis on the spatial and temporal distribution of reservoir quality in the Jurassic Arab D and C members, offshore Abu Dhabi oilfield, UAE. *GeoArabia*, **17**, pp.17–56
- Mountain, G.S., Prell, A.W.L., 1990. A multiphase plate tectonic history of the southeast continental margin of Oman. *Geological Society, London, Special Publications*, p. 725–743, V. **49**(1). doi:[10.1144/GSL.SP.1992.049.01.44](https://doi.org/10.1144/GSL.SP.1992.049.01.44).
- Obermaier, M., Schneider, C., Aigner, T., Claps, M., Von Winterfeld, C., 2010. Paradigm Change: Seal turns into Reservoir. An Outcrop and Modeling Study of a Sudair Formation Equivalent in the Oman Mountains (Jabal Al Akhdar area). *Geo 2010, Poster presentation*.
- Obermaier, M. and Aigner, T., 2012. Field guide for the Triassic Sudair and Jilh Formations (Al Jabal al-Akhdar, Saiq Plateau/Wadi Mu'aydin).
- Osterloff, P., Al-Harthy, A., Penney, R., Spaak, P., Williams, G., Al-Zadjali, F., Jones, N., Knox, R., Stephenson, M.H., Oliver, G. and Al-Husseini M.I. 2004. Depositional sequence of the Gharif and Khuff formations, subsurface Interior Oman. In M.I. Al-Husseini (Ed.), *Carboniferous, Permian and Early Triassic Arabian Stratigraphy*. *GeoArabia Special Publication 3*, Gulf PetroLink, Bahrain, p. 83–147.
- Pöppelreiter, M., Schneider, C. J., Obermaier, M., Forke, H. C., Koehrer, B. and Aigner, T. 2011. Seal turns into reservoir: Sudair equivalents in outcrops, Al Jabal al-Akhdar, Sultanate of Oman. *GeoArabia*, v. **16**, no. 1, p. 69–108.

- Saller, A.H. and Henderson, N., 1998. Distribution of porosity and permeability in platform dolomites: insight from the Permian of West Texas. *AAPG Bull.*, **82**, pp.1528–1550.
- Sharland, P.R., Archer, R., Casey, D.M., Davies, R.B., Hall, S.H., Heward, A.P., Horbury, A.D. and Simmons, M.D. 2001. Arabian Plate Sequence Stratigraphy. *GeoArabia Special Publication 2*.
- Ziegler, M.A. 2001. Late Permian to Holocene Paleofacies Evolution of the Arabian Plate and its Hydrocarbon Occurrences. *GeoArabia*, vol. **6**, pp. 445–504.
- Whittle, G. L. and Alsharhan, A. S. 1995. Observations on the diagenesis of the Lower Triassic Sudair Formation, Abu Dhabi, United Arab Emirates. *Facies*, v. **33**, no. 1, p.185–194.
- Wright, V.P. and Harris, P. 2013. Carbonate Dissolution and Porosity Development in the Burial (Mesogenetic) Environment. *Search and Discovery Article #50860*, *AAPG Annual Convention and Exhibition*, Pittsburgh, Pennsylvania.